

Experiment- 6

Convex lens

AIM:

To find the focal length of a convex lens by plotting the graph between u and v.

APPARATUS:

An optical bench, a convex lens with a lens holder, two optical needles, a knitting needle, and a half-meter scale.

THEORY:

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

Where f is the focal length of the convex lens, u and v are the object and image distances, respectively.

PROCEDURE:

1. Obtain a rough focal length of the convex lens by focusing the image of a distant object on the wall. Twice the value of the focal length gives 2f of the given lens.
2. Mount the given lens on the lens holder and place the upright in position 50cm of an optical bench, which is being levelled with the help of levelling screws and a spirit level.
3. Place the object needle on one side of the lens at a position between f and 2f and the image needle on the other side, as shown in the figure.
4. Remove the parallax between the inverted image of the object needle and the image needle.
5. Measure the distance between the object needle and the optical centre 'OA', note it as u, and the distance of the image needle from the optical centre 'OA', note it as v.
6. Keep moving the object needle by 2 or 3 cm behind '2f' and remove the parallax. Record the respective values as u and v.
7. Now move the object needle 1 to 2 cm in front of the 2f position.
8. Repeat the above steps. Note the values of u and v.
9. Plot the graphs of u against v, with u along the x-axis and v along the y-axis. Draw line OA making 45° with the x-axis and y-axis and find the focal length of the given lens.

RESULT:

The focal length of the given convex lens =

Precautions

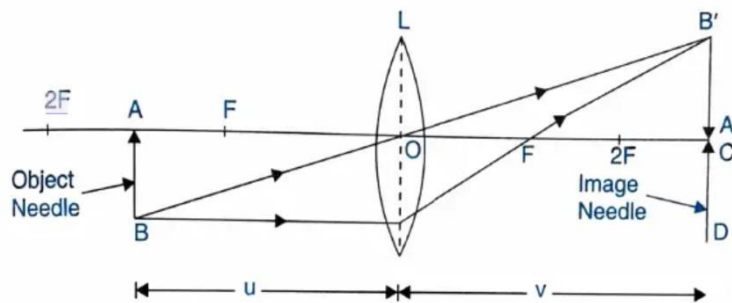
- The tips of the object and image needles should lie at the same height as the centre of the lens.
- Parallax should be removed from tip to tip by keeping the eye at a distance of at least 30cm away from the needle.

Sources of Error

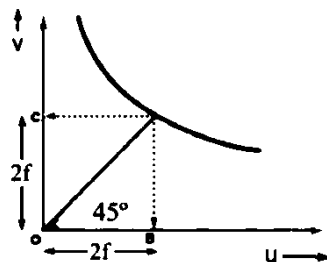
- The upright may not be vertical.
 - Parallax removed may not be perfect.
-

Diagram and observation table on the left side of the page

Ray Diagram:



Model Graph



Observation table

S.No	u(cm)	v(cm)
1		
2		
3		
4		
5		

Calculations:

The focal length of the given convex lens using the formula =

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

The focal length of the given lens using the graph=